Practice · Variables

Each question lists its options; the right one is marked **Right answer**. *Feedback* is what students see after they answer. In a typed question the blank is in square brackets, with the accepted answers separated by `|`.

(mc) L1 · What a variable is

What is a variable?

- A command that prints.
- **Right answer:** A box in memory with a name and a value inside.
- A file ending in `.c`.
- A kind of error.

Feedback: A variable is a named box. `int age = 20;` makes a box called `age` with 20 in it.

(mc) L1 · The type for a whole number

Which type holds a whole number, like 20 or -3?

- `double`
- `char`
- **Right answer:** `int`
- `number`

Feedback: `int` is for whole numbers. `double` is for numbers with a decimal point. `char` is for one character. `number` is not a C type.

(mc) L1 · The type for a price

Which type would you use for a price like 4.75?

- `int`
- **Right answer:** `double`
- `char`
- `price`

Feedback: 4.75 has a decimal point, so `double`. An `int` would throw the .75 away.

(mc) L2 · What = means

In C, what does `x = x + 1;` do?

- It checks whether x equals x + 1.
- It is an error, because no number equals itself plus one.
- **Right answer:** It calculates x + 1 and stores the result back in x.
- It prints x + 1.

Feedback: `=` is not "equals". It means "work out the right side, then put the answer in the box on the left". If x was 5, it becomes 6.

(mc) L2 · Trace it

What does this print?

```
int y = 3;
```

```
y = y * 2 + 1;
printf("%d\n", y);
```

- `3`
- `8`
- `6`
- **Right answer:** `7`

Feedback: Right side first: y is 3, so 3 × 2 = 6, then 6 + 1 = 7. Then 7 is stored in y. Multiplication comes before addition, as in school. 8 is what you get if you add first, which is wrong.

(mc) L2 · Single or double quotes

Which line is correct?

- **Right answer:** `char grade = 'A';`
- `char grade = "A";`
- `char grade = A;`
- `char grade = 'A"`

Feedback: One character goes in single quotes: `'A'`. Double quotes are for text. Without any quotes, `A` would be taken as the name of a variable.

(fib) L3 · Declare a whole number

Write a line that makes a variable called `count` for whole numbers and puts 5 in it: [int count = 5;|int count=5;]

Feedback: `int count = 5;`. Type, name, equals, value, semicolon.

(fib) L3 · Declare a decimal number

Write a line that makes a variable called `price` with a decimal point and puts 2.5 in it: [double price = 2.5;|double price=2.5;]

Feedback: `double price = 2.5;`. A number with a decimal point needs `double`.

(fib) L3 · Add one

Write one line that adds 1 to the variable `x` that already exists: [x = x + 1;|x=x+1;|x += 1;|x+=1;|x++;]

Feedback: `x = x + 1;`. Read it as: take x, add one, put the answer back in x.